GenAI. Lecture 4 - StyleGAN

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Recap

Recap. Key points

- GAN is unstable, but really fast in practice
- Wasserstein distance helps to overcome GANs gradient vanishing
- Several ways to satisfy Lipshitz constraints

StyleGAN

Introduction

StyleGAN is an improved GAN architecture that introduces:

- Style-based generation via Adaptive Instance Normalization (AdaIN)
- An intermediate latent space for better disentanglement
- Stochastic variation for fine details
- Newly introduced metrics for latent space evaluation
- A new dataset: Flickr-Faces-HQ (FFHQ)

Limitations of Traditional GANs

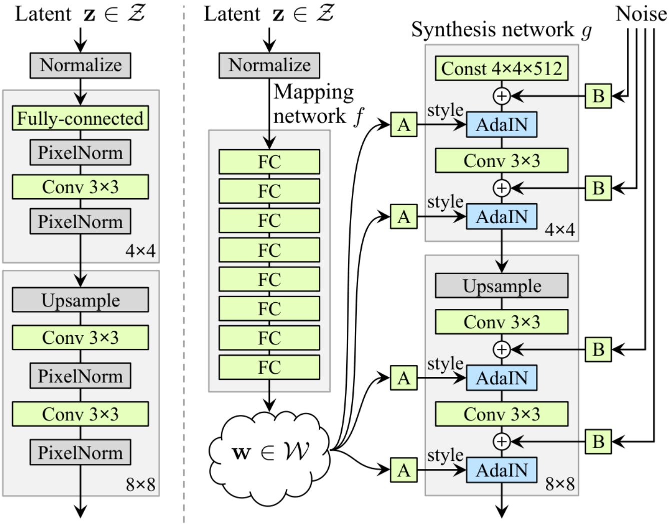
- Latent space entanglement. Difficult to control specific attributes
- Features are not learned hierarchically
- Lack of control over fine details
- Global transformations affect the entire image

StyleGAN Architecture

- Uses an **intermediate latent space** W instead of Z
- Latent code is passed through a **mapping network**
- Feature maps are modulated using **Adaptive Instance Normalization (AdaIN)**
- Stochastic noise is injected to introduce fine-grained variations

Mapping Network

- Traditional GANs map latent vector z directly to images
- StyleGAN introduces a **mapping network** $f: Z \to W$
- This allows control over different levels of detail in the image



Adaptive Instance Normalization (AdaIN)

- Each layer's feature maps are modulated using a learned style
- Equation:

$$\mathrm{AdaIN}(x_i, y) = y_{s,i}\frac{x_i - \mu(x_i)}{\sigma(x_i)} + y_{b,i}$$

- Provides control over different aspects of the image

Style Mixing Regularization

- Regularization technique to improve latent space disentanglement
- Different style vectors control different layers in the generator
- Forces the network to generalize style attributes instead of memorizing correlations
- Helps prevent dependencies between features at different scales

Stochastic Variation

- StyleGAN injects per-pixel Gaussian noise at different scales
- Controls fine details such as hair strands, freckles, and skin texture
- Separates global attributes (e.g., pose) from local attributes (e.g., texture)

$$x_i = x_i + B_i\, n_i$$

$$n_i \sim \mathcal{N}(0,1)$$

B_i is learnable

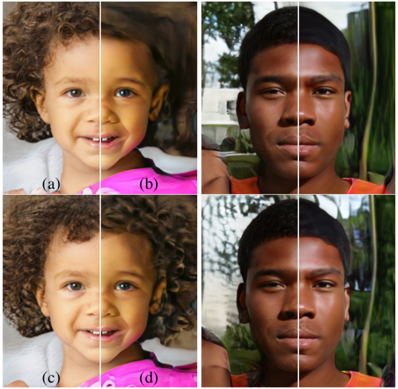
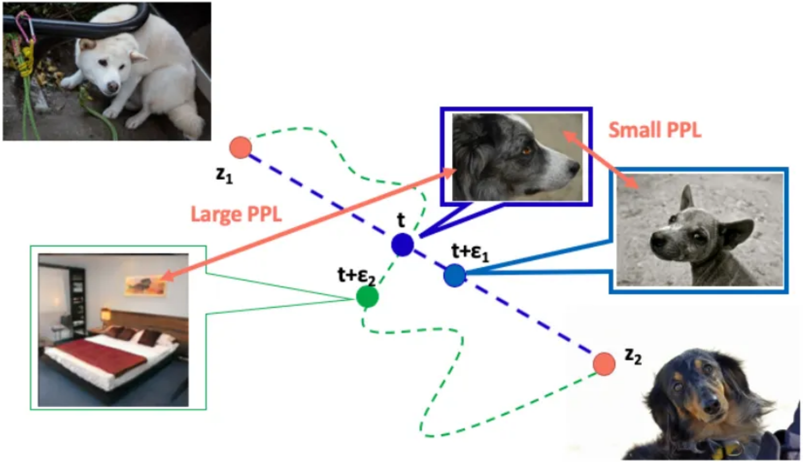


Figure 5. Effect of noise inputs at different layers of our generator. (a) Noise is applied to all layers. (b) No noise. (c) Noise in fine layers only (64^2 – 1024^2). (d) Noise in coarse layers only (4^2 – 32^2). We can see that the artificial omission of noise leads to featureless "painterly" look. Coarse noise causes large-scale curling of hair and appearance of larger background features, while the fine noise brings out the finer curls of hair, finer background

Newly Introduced Metrics

- **Learned Perceptual Path Length (LPPL)**: Measures smoothness of latent space transitions $l_Z = \mathbb{E}\left[\frac{1}{\epsilon^2} d\left(G(\mathrm{slerp}(\mathbf{z}_1, \mathbf{z}_2; t)), G(\mathrm{slerp}(\mathbf{z}_1, \mathbf{z}_2; t+\epsilon))\right)\right]$
- **Linear Separability**: Evaluates disentanglement of latent factors (SVM classifier on latent space $\mathcal{W}$)
- **FID Score**: Measures realism of generated images

Learned Perceptual Length



Results and FID Score

- StyleGAN achieves lower Fréchet Inception Distance (FID) than previous GANs
- Produces highly realistic, high-resolution images

Limitations Future Work

- Artifacts: Some outputs contain "water droplet" artifacts
- Latent space still has some entanglement
- Improvements in StyleGAN2 address these issues

Conclusion

- StyleGAN introduces a novel way to control generative models
- Disentanglement via style-based generation improves realism
- Stochastic variation helps produce diverse, high-quality outputs

Future Work: StyleGAN2, improved regularization techniques.

StyleGAN-2

Introduction

StyleGAN2 improves upon StyleGAN by introducing:

- Removal of normalization artifacts (water droplet effects)
- Better generator regularization (Path Length Regularization)
- Replacing progressive growing with skip and residual networks
- Enhanced latent space structure for better interpolations

Limitations of StyleGAN1

- ”Water droplet” artifacts due to AdaIN normalization
- Disentanglement issues in latent space
- Progressive growing caused feature sticking and phase artifacts
- Inefficient high-resolution feature usage

StyleGAN2 Architecture Improvements

- **Weight Demodulation:** Replaces adaptive activations (AdaIN) to adaptive weights to remove artifacts
- **Skip Connection Generator:** Avoids phase artifacts
- **Residual Discriminator:** Stabilizes training
- **Better High-Resolution Feature Utilization:** Efficiently distributes details

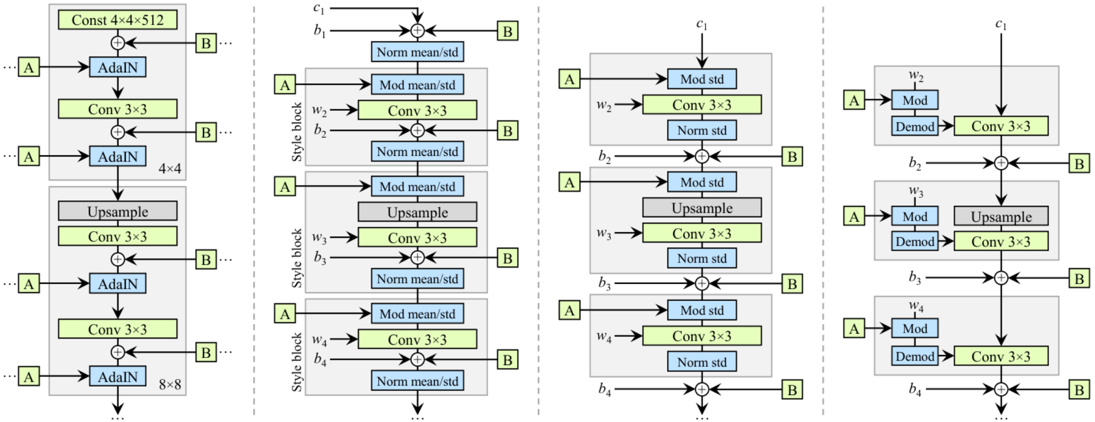
Weight Demodulation

- AdaIN was replaced to prevent feature correlations causing artifacts
- Normalization applied directly to weights, not feature maps
- Prevents "water droplet" patterns from emerging

$$\tilde{\mathbf{w}}_{i,j,k} = s_j \cdot \mathbf{w}_{i,j,k}$$

$$\hat{\mathbf{w}}_{i,j,k} = \frac{\tilde{\mathbf{w}}_{i,j,k}}{\sqrt{\sum_{i,k} \tilde{\mathbf{w}}_{i,j,k}^2 + \epsilon}}$$

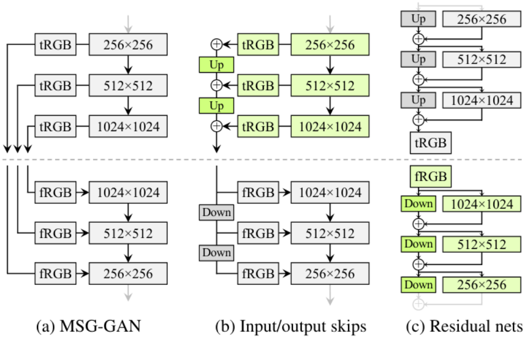
StyleGAN2 architecture



(a) StyleGAN (b) StyleGAN (detailed) (c) Revised architecture (d) Weight demodulation

Skip and Residual Connections

- Avoids phase artifacts caused by progressive growing
- Skip connections ensure all resolutions contribute equally
- Residual learning stabilizes the discriminator



Path-Length Regularization

- Ensures smooth latent space transitions
- Penalizes variations in perceptual distance for interpolations
- Encourages more linear transformations in the generator

$$\mathbb{E}_{\mathbf{w},\mathbf{y}\sim\mathcal{N}(0,\mathbf{I})}\left(\left\|\mathbf{J}_{\mathbf{w}}^{T}\mathbf{y}\right\|_2 - a\right)^2$$

R1

- Regularization applied to the discriminator to prevent overfitting
- Penalizes the squared gradient norm of the discriminator on real images
- Helps stabilize training and encourages smooth discriminator gradients
- Defined as:

$$R_1 = \frac{\gamma}{2}\mathbb{E}_{x\sim p_{\text{data}}}\left[\|\nabla D(x)\|^2\right]$$

Lazy Regularization

- Regularization is applied less frequently to reduce computational overhead
- Instead of applying regularization every step, it is applied periodically
- Maintains training stability while improving efficiency
- Used in path-length regularization and R1 gradient penalty

Results and Improvements

- Lower FID scores than StyleGAN1
- No more water droplet artifacts
- More stable training without progressive growing
- Smoother latent space interpolations

Configuration	FFHQ, 1024×1024				LSUN Car, 512×384			
	FID ↓	Path length ↓	Precision ↑	Recall ↑	FID ↓	Path length ↓	Precision ↑	Recall ↑
A Baseline StyleGAN [24]	4.40	212.1	**0.721**	0.399	3.27	1484.5	**0.701**	0.435
B + Weight demodulation	4.39	175.4	0.702	0.425	3.04	862.4	0.685	0.488
C + Lazy regularization	4.38	158.0	0.719	0.427	2.83	981.6	0.688	0.493
D + Path length regularization	4.34	**122.5**	0.715	0.418	3.43	651.2	0.697	0.452
E + No growing, new G & D arch.	3.31	124.5	0.705	0.449	3.19	471.2	0.690	0.454
F + Large networks (StyleGAN2)	**2.84**	145.0	0.689	**0.492**	**2.32**	**415.5**	0.678	**0.514**
Config A with large networks	3.98	199.2	0.716	0.422	–	–	–	–

Limitations and Future Work

- Still some entanglement in latent space
- High computational cost for training
- Improvements needed in conditioning and texturing

Conclusion

- StyleGAN2 refines GAN image synthesis with better architectures
- Removal of artifacts and improved regularization techniques
- Future improvements can enhance realism and controllability

Future Work: StyleGAN3 and beyond

StyleGAN-3

Introduction

StyleGAN3 improves upon StyleGAN2 by introducing:

- Alias-free convolutional networks
- Equivariance for translation and rotation
- Fourier-based continuous signal interpolation
- Removal of explicit noise injection

Limitations of StyleGAN2

- Aliasing artifacts due to fixed-grid processing
- Feature sticking and spatial inconsistencies
- Poor equivariance to translation and rotation
- Image transformations were not smooth

StyleGAN2 artifacts

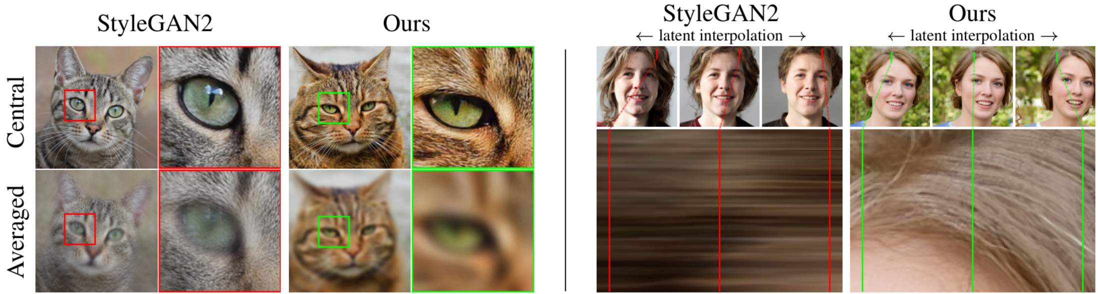


Figure 1: Examples of "texture sticking". **Left:** The average of images generated from a small neighborhood around a central latent (top row). The intended result is uniformly blurry because all details should move together. However, with StyleGAN2 many details (e.g., fur) stick to the same pixel coordinates, showing unwanted sharpness. **Right:** From a latent space interpolation (top row), we extract a short vertical segment of pixels from each generated image and stack them horizontally (bottom). The desired result is hairs moving in animation, creating a time-varying field. With StyleGAN2 the hairs mostly stick to the same coordinates, creating horizontal streaks instead.

Discretization problems

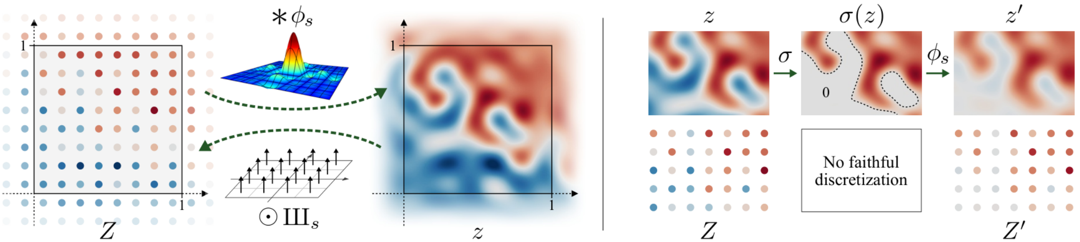


Figure 2: **Left:** Discrete representation Z and continuous representation z are related to each other via convolution with ideal interpolation filter ϕ_s and pointwise multiplication with Dirac comb $Ш_s$. **Right:** Nonlinearity σ, ReLU in this example, may produce arbitrarily high frequencies in the continuous-domain $\sigma(z)$. Low-pass filtering via ϕ_s is necessary to ensure that Z' captures the result.

Introduction

In **StyleGAN3**, images are treated as **continuous signals** rather than discrete pixel grids.
This allows **Fourier-based transformations** such as:

- Smooth translations
- Rotations
- Scaling

This presentation breaks down the process step by step.

What is the Fourier Transform of an Image?

A **Fourier Transform (FT)** converts an image from the **spatial domain** (pixels) to the **frequency domain** (sinusoids of different frequencies).

Benefits:

- Enables direct transformations (translation, rotation, scaling) in frequency domain.
- Provides smooth, alias-free modifications.

2D Fourier Transform Formula

The **2D Discrete Fourier Transform (DFT)** for an image $I(x, y)$ is:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y)\, e^{-i2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)} \tag{1}$$

Where:

- $I(x, y)$ is the **pixel intensity** at position (x, y).
- $F(u, v)$ is the **Fourier coefficient** at frequency (u, v).
- The exponent represents **sinusoidal basis functions**.

Fourier Transform Property: Translation

If an image is shifted by $(\Delta x, \Delta y)$, the Fourier Transform property states:

$$F'(u,v) = F(u,v)e^{-i2\pi(u\Delta x + v\Delta y)} \tag{2}$$

Effect: Smooth translation in image space corresponds to a phase shift in the frequency domain.

Fourier Transform Property: Rotation

To rotate an image by an angle θ, we rotate its frequency coefficients:

$$F'(u, v) = F(u', v') \tag{3}$$

Where:

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} \tag{4}$$

Effect: Rotation in the spatial domain corresponds to rotating the frequency domain representation.

Fourier Transform Property: Scaling

To scale an image by a factor s:

$$F'(u,v) = \frac{1}{s^2} F\left(\frac{u}{s}, \frac{v}{s}\right) \tag{5}$$

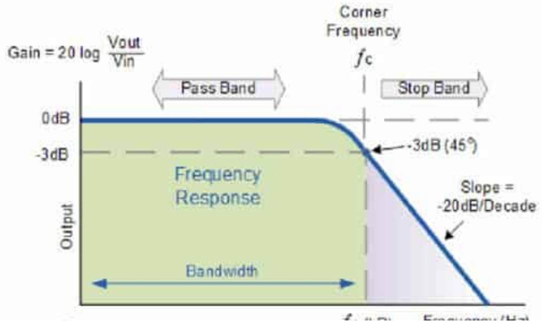
Effect: Scaling the image compresses or expands its frequency representation.

Applying Filtering in Frequency Domain

Filtering in the frequency domain is done by multiplying the Fourier-transformed image by a filter $H(u, v)$:

$$F'(u, v) = F(u, v) \cdot H(u, v) \tag{6}$$

Effect: This allows for operations like **low-pass filtering** to remove high-frequency noise.



Continuous Representation Using Whittaker-Shannon Formula

Given **discrete** $Z[x]$ (with sampling rate s) of a **continuous** $z(x)$, the **Whittaker-Shannon interpolation formula** states:

$$z(x) = (\phi_s * Z)(x) \tag{7}$$

Where:

- $\phi_s(x, y) = \mathrm{sinc}(sx) \cdot \mathrm{sinc}(sy)$
- $\mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$

ϕ_s has a **bandlimit of** $s/2$ along both dimensions, ensuring alias-free signal representation.

Conversion from Continuous to Discrete Domain

Conversion from the continuous to discrete domain corresponds to sampling $z(x)$ at points of $Z[x]$.

$$\mathbb{I}_s(x) = \sum_{X \in \mathbb{Z}^2} \delta\left(x - \frac{X + \frac{1}{2}}{s}\right) \tag{8}$$

Effect: This places the sample points at **pixel centers**, aligning the continuous representation with the discrete grid.

Signal Processing Formulas in StyleGAN3

The following equations describe signal processing steps in StyleGAN3:

$$\mathbf{f}(z) = \phi_{s'} * \mathbf{F}(\mathbb{I}_s \odot z) \tag{9}$$

$$\mathbf{F}(Z) = \mathbb{I}_{s'} \odot \mathbf{f}(\phi_s * Z) \tag{10}$$

Where:

- ϕ_s is the interpolation filter.
- $\mathbb{I}_s$ is the sampling operator.
- $\odot$ represents element-wise multiplication.

Introduction to Equivariant Network Layers

In **StyleGAN3**, operations are designed to be **equivariant** with respect to spatial transformations like **translation** and **rotation**.

- Equivariance ensures that transformations in the input lead to corresponding transformations in the output.
- Inputs are bandlimited to $s/2$, ensuring no frequency content exceeds this limit.

Types of Equivariance

StyleGAN3 focuses on two types of equivariance:

- **Translation Equivariance**: The model output shifts correspondingly when the input is shifted.
- **Rotation Equivariance**: Rotating the input rotates the output, with stricter spectral constraints.

Spectral Constraint for Rotation:

- The spectrum is limited to a disc with radius $s/2$.
- Applies to initial network input and downsampling filters.

Primitive Operations in Equivariant Networks

Key operations in a generator network include:

- **Convolution**
- **Upsampling**
- **Downsampling**
- **Nonlinearity**

Note: Linear combinations of features do not affect equivariance.

Convolution in Equivariant Networks

Consider a standard convolution with a discrete kernel K.

$$\mathbf{F}_{\text{conv}}(Z) = K * Z \tag{11}$$

The corresponding continuous operation is:

$$\mathbf{f}_{\text{conv}}(z) = \phi_s * (K * (\mathbb{I}_s \odot z)) = K * z \tag{12}$$

Key Points:

- Convolution commutes with translation in the continuous domain.
- For rotation equivariance, K must be radially symmetric.

Upsampling in Equivariant Networks

Ideal upsampling does not modify the continuous representation.

- Purpose: Increase the output sampling rate $s' > s$.
- Maintains translation and rotation equivariance.

Continuous domain operation:

$$\mathbf{f}_{\mathrm{up}}(z) = z \tag{13}$$

Discrete operation:

$$\mathbf{F}_{\mathrm{up}}(Z) = \mathbb{I}_{s'} \odot (\phi_s * Z) \tag{14}$$

Implementation of Upsampling

For $s' = ns$ with integer n:

- Interleave Z with zeros to increase its sampling rate.
- Convolve with a discretized filter $\mathbb{I}_{s'} \odot \phi_s$.

Result: Smooth upsampling without introducing new frequencies.

Downsampling in Equivariant Networks

Downsampling requires low-pass filtering to remove frequencies above the output bandlimit. Continuous domain operation:

$$\mathbf{f}_{\text{down}}(z) = \psi_s * z \tag{15}$$

Where $\psi_s = \psi * \phi_s$ is an ideal low-pass filter normalized to unit mass.

Downsampling Filters and Properties

The downsampling filter ψ_s is defined as:

$$\psi_s := s^2 \cdot \phi_s \tag{16}$$

Filter composition property:

$$\psi_s * \psi_{s'} = \psi_{\min(s,s')} \tag{17}$$

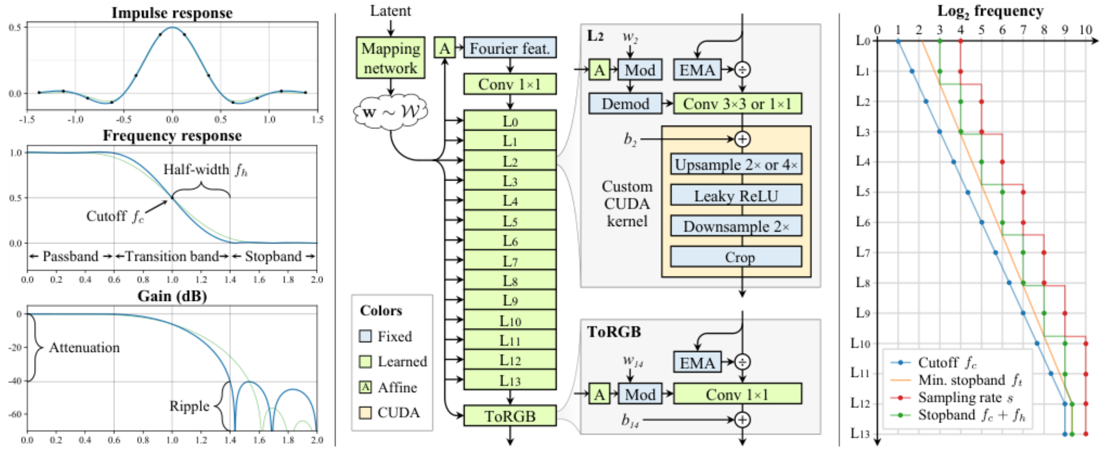
Jinc Function and Radial Filters

The ideal radial filter is expressed using the **jinc** function:

$$\phi_s^{\circ}(x) = \mathrm{jinc}(s\|x\|) = \frac{2J_1(\pi s\|x\|)}{\pi s\|x\|} \tag{18}$$

Where J_1 is the first-order Bessel function of the first kind.

StyleGAN3 architecture



(a) Filter design concepts (b) Our alias-free StyleGAN3 generator architecture (c) Flexible layers

Input Fourier Features in StyleGAN3

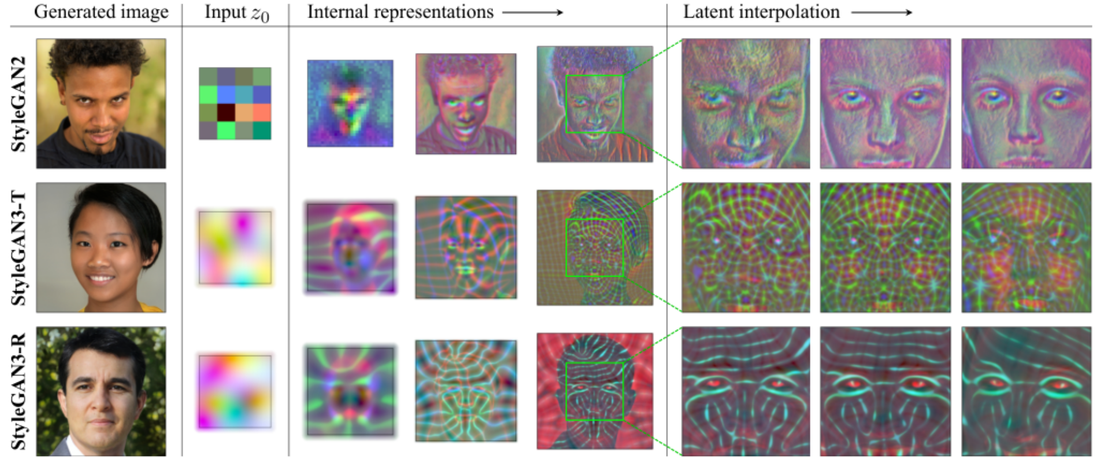
StyleGAN3 replaces the learned constant input from previous versions with **Fourier features**. These features encode spatial information continuously, ensuring equivariance to transformations.

- **Fourier Features Definition:**

$$\Phi(x, y) = [\sin(2\pi(k_x x + k_y y)), \cos(2\pi(k_x x + k_y y))] \tag{19}$$

- k_x, k_y are sampled frequencies that define the sinusoidal patterns.
- These features allow smooth translations and rotations in the latent space.

StyleGAN3 results



Conclusion

- **Equivariant Layers:** StyleGAN3 introduces equivariant layers that maintain translation and rotation consistency, crucial for generating high-quality images.
- **Frequency Management:** The architecture uses bandlimited inputs and low-pass filters to prevent aliasing and ensure faithful signal representation.
- **Convolution, Upsampling, Downsampling:** All operations are carefully designed to preserve the continuous nature of images, using techniques like jinc-based radial filters.
- **Impact:** These advancements result in smoother, more consistent outputs that respond naturally to geometric transformations.